

Down the Rabbit Hole:

Experiencing Alice in Wonderland Syndrome through Virtual Reality

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ABSTRACT

Alice in Wonderland Syndrome (AIWS) is a rare perceptual disorder affecting visual processing, the perception of one's body and the experience of time. This condition can be congenital or result from various insults to the brain. There is growing interest in AIWS in providing a window into how different areas of the brain work together to construct reality. We developed a virtual reality (VR) simulation of this condition as a psychoeducational tool for students in the psychological and medical sciences and care givers to experience the different perceptual distortions common in AIWS and an opportunity to reflect on the nature of perception.

CCS CONCEPTS

• Applied Computing; • Law, social and behavioral science; • Psychology;

KEYWORDS

Perception, Alice in Wonderland Syndrome, Education, Virtual Reality

ACM Reference Format:

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1 INTRODUCTION

ACM's Alice in Wonderland Syndrome (AIWS) is a relatively obscure neuropsychological condition characterized by changes in visual processing, bodily perception and the experience of time. First documented in the 1950s [Todd, 1955], this condition was named for Alice in Lewis Carroll's *Alice in Wonderland* whose experiences in the story mirror the perceptual distortions seen in



Figure 1: Illustrations from Lewis Carroll's *Alice in Wonderland* with Alice experiencing changes in bodily perception (<https://digitalcollections.nypl.org/> Creative Commons CC0 1.0)

the condition [Carroll, 1865; see Fig. 1], and since then only about ~170 cases have been described in the medical literature [Blom, 2016].

Visual phenomena include seeing objects as larger or smaller than they really are while changes in bodily perception include experiencing parts or the whole body as being larger or smaller than in reality. The perception of time can also be altered with time seeming to speed up or slow down. These phenomena are generally unpredictable, of short duration, and may happen up to several times a day. AIWS is more common in children with up to 30% teenagers experiencing brief episodes of AIWS symptoms. Different conditions that have been associated with AIWS include migraine, temporal lobe epilepsy, brain tumor, viral infections, and psychoactive drugs [Blom, 2016]. Neuroimaging data suggest changes in the activity and functional connectivity of visual and attentional networks in the brain of individuals experiencing this condition [Brumm et al., 2010, Piervincenzi et al., 2023]. The relative obscurity of AIWS and the rather fantastical nature of symptoms are fertile ground for a compelling immersive VR experience to both educate students and clinicians about this condition and the highly constructive nature of perception.

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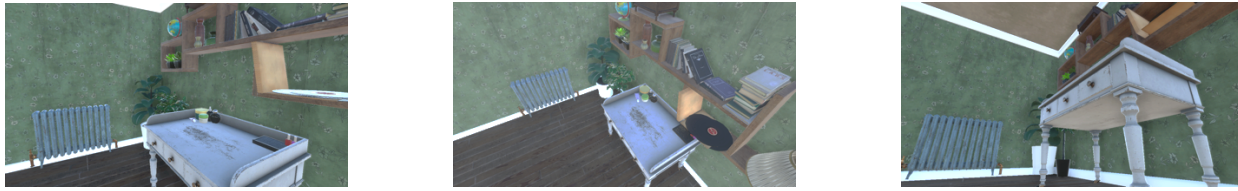


Figure 2: The first image is a normal perspective of the room where the second and third images illustrate the changes in perspective as the result of macrosomatognosia and microsomatognosia respectively.

2 IMPLEMENTATION

To create our virtual simulation, we consulted multiple sources including scientific case studies, YouTube videos from individuals with AIWS as well as conversations with a former lab member with AIWS to identify some of the most common perceptual changes in AIWS and how they may manifest in daily life. From these we chose three primary experiences 1) macropsia/micropsia in which objects seem to grow or shrink 2) macrosomatognosia/microsamatognosia in which the body grows or shrinks and 3) quick-motion/protracted duration in which the perception of time speeds up or slows down. Using the Unity game engine, we designed a simulation where users experience these symptoms firsthand in a real-world setting, a residential house. Users start in the living room where they watch a short, animated video on a television screen providing an overview of AIWS and the types of perceptual distortions the user can expect in the simulation. Teleport is then used to allow users to travel to different parts of the house as well as the backyard. The symptoms of macropsia/micropsia take place in the kitchen and dining room area with items such as pots and appliances randomly growing or shrinking as well as a painting on the wall. Macrosomatognosia and microsamatognosia occur in a bedroom where the user grows and shrinks inside the room (See Fig. 2).

Partial macrosomatognosia occurs with a hand model attached to the controller where the fingers grow large and then shrink back to normal. The final symptom of AIWS is time distortion. Here, the user is teleported to the back yard of the house where, in a nod to Carroll's story, they interact with a white rabbit who randomly speeds up and slows down. While we have received positive feedback from our colleague with AIWS on the veracity of our simulation, additional feedback from more individuals with AIWS would be helpful in future iterations of this project.

3 CONCLUSION

Virtual reality has become a popular tool in neuropsychological assessment and rehabilitation [Kourtesis et al., 2020], but its use in educating students and practitioners through first person experience of clinical symptoms is currently underutilized. There is evidence that using VR in this fashion not only increases an individual's knowledge of a neuropsychological condition, but also their empathy with the result of decreasing stigma towards people who experience the world differently [Zare-Bidaki et al., 2022]. In our future work, we will test how our simulation affects neurotypical people's understanding of how reality is constructed by the senses.

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